

Ecology & Ecosystems

No living thing lives alone. Every plant and animal needs others — for food, shelter, and energy. This guided edition walks you through how living things connect, how energy and matter move, and the big biomes of Earth — step by step.

 Georgia Standard S7L4

WHAT YOU'LL LEARN

01 Parts of an Ecosystem

02 Jobs Organisms Do

03 Food Webs

04 Energy Flow

05 Cycling of Matter

06 Limiting Factors

07 Earth's Biomes

08 Review & Check



CHAPTER WORD BANK

ecosystem

biotic

abiotic

producer

consumer

decomposer

food web

energy pyramid

limiting factor

biome

The  boxes explain each word.

4.1 Parts of an Ecosystem

An **ecosystem** is all the living and non-living things in one area, and how they work together. A pond is an ecosystem. A forest is too.

biotic = the LIVING parts (plants, animals).

abiotic = the NON-LIVING parts (water, sunlight, air, soil).



Biotic (living)

Plants, animals, fungi, bacteria.



Abiotic (non-living)

Sunlight, water, air, soil, rocks.

✓ Quick Check 4.1

1. Is a fish biotic or abiotic?
2. Is sunlight biotic or abiotic?

4.2 Jobs Organisms Do

Every living thing has a **job** based on how it gets energy. There are three main jobs.

Job	How It Gets Energy	Example
Producer	Makes its own food from sunlight	Plants
Consumer	Eats other living things	Rabbit, fox
Decomposer	Breaks down dead things	Fungi, bacteria

Three Kinds of Consumers

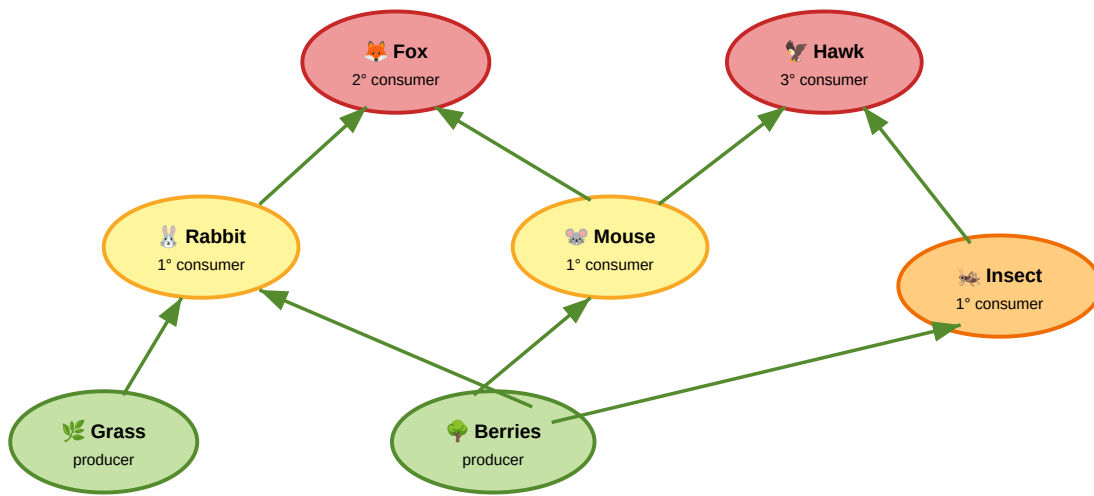
Herbivore = eats only plants. **Carnivore** = eats only animals. **Omnivore** = eats both.

✓ Quick Check 4.2

1. What job makes its own food from sunlight?
2. A bear eats plants AND animals. What kind of consumer is it?

4.3 Food Webs

A **food web** shows how energy moves from one living thing to another. The arrows show where the energy goes.



Arrows point from food → to the eater (the direction energy flows)

Figure 4.1 — A food web. Each arrow points to the eater.

★ Arrow Rule

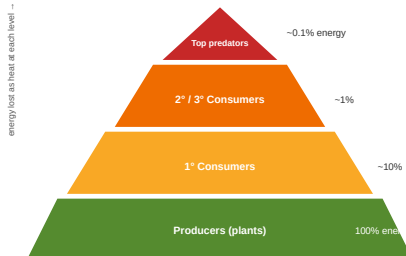
The arrow points FROM the food TO the eater. Grass → Rabbit means the rabbit eats the grass and gets its energy.

🔪 Sentence Starter

In a food web, the arrows point toward _____.

4.4 Energy Flow

Energy comes from the sun. Producers catch it. Then it moves up — but most energy is LOST at each step.



An **energy pyramid** shows energy getting smaller as you go up. Only about **10%** moves to the next level.

That is why there are MANY plants but only a FEW top predators.

★ The 10% Rule

Only 10% of energy moves up each step. Grass 10,000 → Rabbit 1,000 → Fox 100 → Hawk 10.

✓ Quick Check 4.4

1. About what percent of energy moves to the next level?
2. Are there more producers or more top predators?

4.5 Cycling of Matter

Energy **flows** and gets lost. But matter **cycles** — it is used over and over. Water and carbon go round and round.

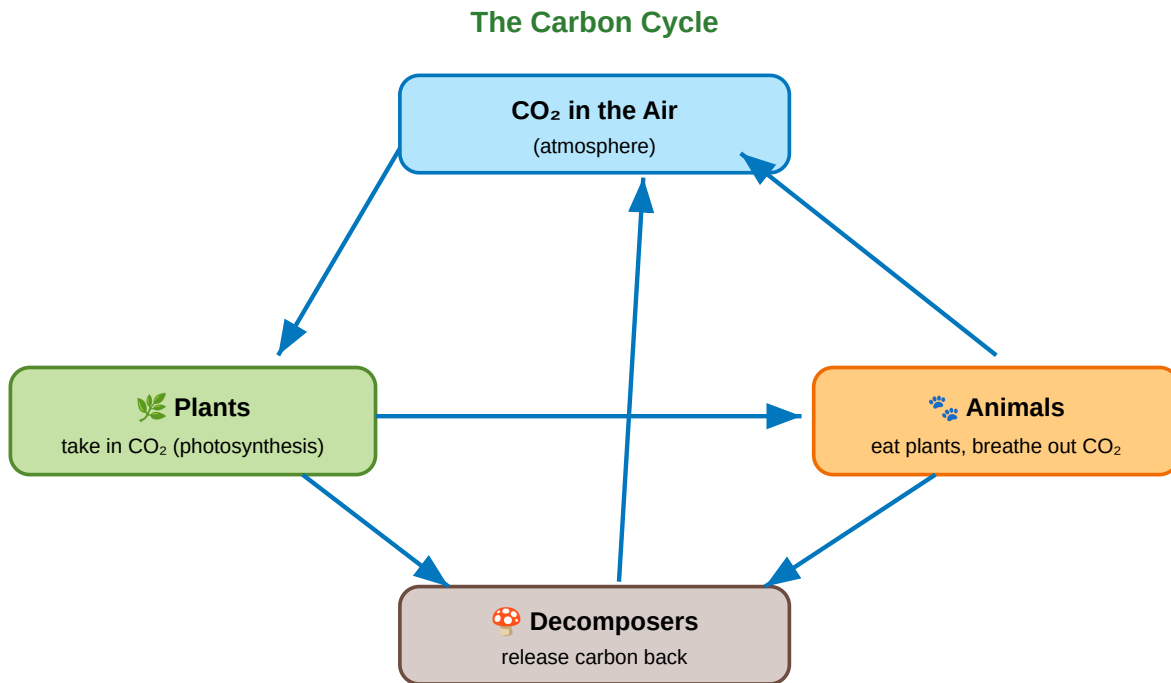


Figure 4.2 — The carbon cycle. Carbon moves between air, living things, and soil.

flow = moves one way and is lost (energy).

cycle = goes round and round, reused (matter).

🔑 Big Difference

Energy **flows** one way. Matter **cycles** forever.

4.6 Limiting Factors

A **limiting factor** is anything that stops a population from growing too big. It keeps nature in balance.

Limiting Factor	What It Does
Resources	Not enough food, water, or space
Disease	Sickness spreads and lowers numbers
Climate	Droughts, floods, or extreme cold/heat
Humans	Pollution, building, and hunting

✓ Quick Check 4.6

1. Name one limiting factor that could lower a population.

4.7 Earth's Biomes

A **biome** is a big area with its own weather, plants, and animals. Here are the main ones.

 Rainforest Hot & wet	 Savanna Warm, grassy	 Forest 4 seasons	 Desert Very dry
 Grassland Grasses	 Taiga Cold forest	 Tundra Frozen	 Aquatic Water

🔑 Water Ecosystems

Freshwater = low salt (lakes, rivers). Estuary = where river meets ocean. Marine = salty ocean.

4.8 Review & Check

Great job! Here is everything in one place.

Big Idea	Remember This
Ecosystem parts	Biotic (living) + abiotic (non-living).
Jobs	Producers make food, consumers eat, decomposers recycle.
Energy flow	Only ~10% moves up each step.
Matter	Water and carbon cycle forever.
Limiting factors	Resources, disease, climate, humans.
Biomes	7 land biomes + water ecosystems.

★ Standard Check — S7L4

You can now **(a)** explain how living things interact, **(b)** show energy flow and matter cycling, **(c)** tell how factors affect populations, and **(d)** name Earth's biomes. 🎉